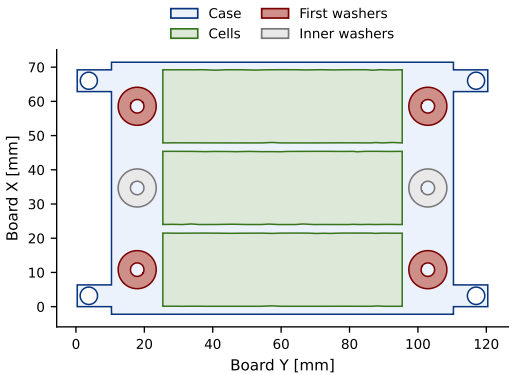
The outer washers meet the case before the cells



Plan view through the cover. Red marks the first four loaded washers.

Lowering the case from 3 mm

The four outer washers touch after 0.603 mm of travel. The rigid tabs are still 2.397 mm from the PCB. The first cell touches after 1.760 mm.

Flush installation

With nylon screws, the flexible-board model crossed the PLA through-layer reference at 0.95–1.10 mm of travel. We stopped at 2.75 mm. All four tabs touched, with only 4.4–5.1% of each footprint within 0.05 mm of the PCB. The cells still had clearance.

The hole pattern leaves little room for an M4 screw

	PCB	Case
Bore diameter [mm]	4.200	5.000–5.090
Hole pitch X [mm]	62.865	62.855
Hole pitch Y [mm]	114.427	113.241

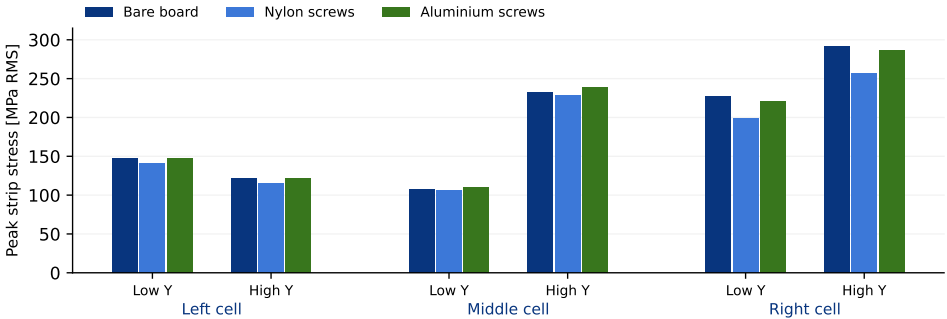
Each end of the centred pattern is about 0.593 mm off-axis. The smaller bore pair allows 0.600 mm with a nominal M4 shank.

Hardware under study

Four M4 case screws in aluminium or nylon. The nylon reference is McMaster-Carr 93640A145.

The dense PLA case adds about 74 g to a 252 g bare assembly. Print orientation and infill affect its installed stiffness and mass.

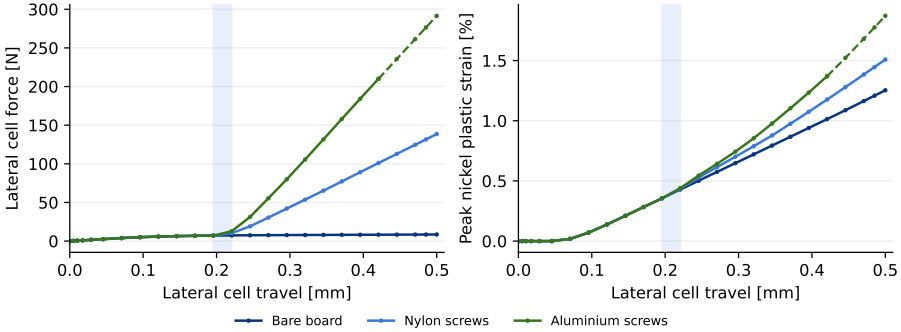
The raised nylon case lowers vibration stress in all six nickel strips



With base X excitation, the highest strip stress was **291.9 MPa RMS** bare, **256.2** with nylon screws and **286.5** with aluminium. All three maxima occurred at the same corner of the right-hand cell attachment.

In the seeded history, all three elastic models crossed the local pull study's assumed 185 MPa yield stress within 39 ms. Both raised cases still had clearance at the cells. With base Z, the largest lengthwise force at a right-cell attachment rose from 2.07 N RMS bare to 4.14 N with nylon and 3.55 N with aluminium.

Both raised cases limit cell travel under the same pull

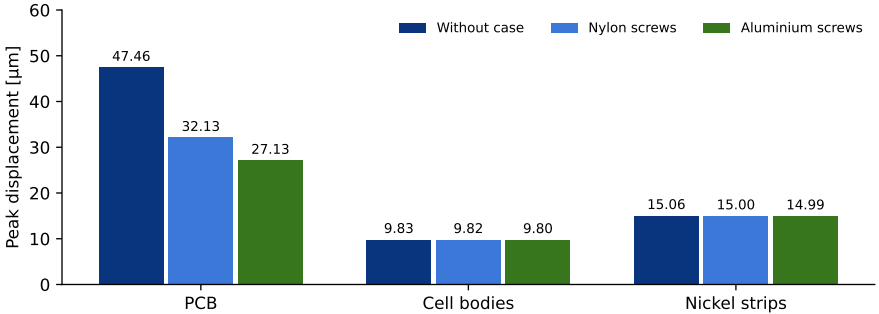


At **8.54 N**, the bare cell travelled 0.50 mm. Both raised cases travelled about 0.20 mm. Their difference was smaller than the spacing between saved states.

At the same **0.421 mm travel**, the required pull was 8.26 N bare, 101.0 N with nylon screws and 210.0 N with aluminium. The aluminium case crossed the first printed-PLA tensile reference in the next increment.

Positive X edge-cell pull at 3 mm stand-off. The local fixture holds the case screw tops fixed.

Aluminium screws further reduced PCB deflection during tightening

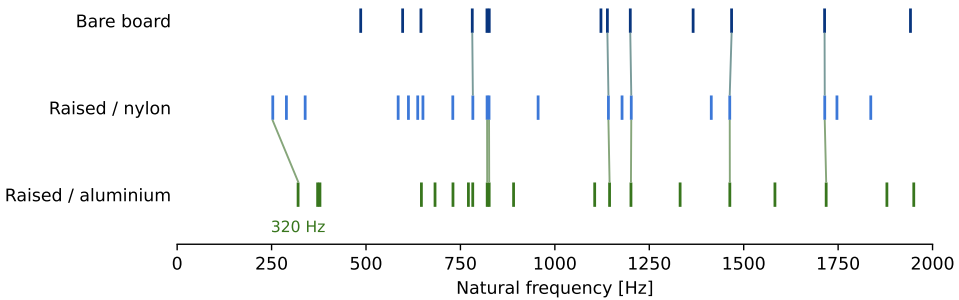


The same PCB node moved 47.46 μm bare, 32.13 μm with nylon screws and **27.13 μm** with aluminium screws. The reductions from bare were 32% and 43%, respectively.

No cell, strip or washer contacted the case during tightening. Load transfer was through the four corner attachments.

3 mm stand-off. Six original bolts loaded to 750 N each and locked. No imposed vibration.

Both raised cases introduce modes below 486 Hz

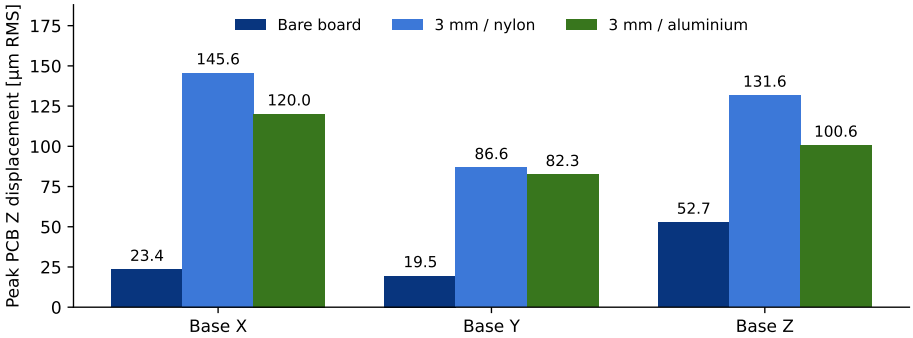


The first cased mode rises from **253 Hz with nylon** to **320 Hz with aluminium**. Its shape remains closely matched across 399,194 solid nodes (MAC 0.989).

The closest match to bare mode 20 (2364 Hz) is mode 27 in both raised assemblies, at 2365 Hz with nylon and 2368 Hz with aluminium. With the original unpreloaded contacts, mode 1 stayed at 66.61 Hz in all three assemblies.

Bolt-locked flexible assemblies at 3 mm stand-off. Undamped mode shapes about the tightened state.

The open-gap vibration model predicts more PCB motion



With base X excitation, peak PCB Z motion was **23.37** μm RMS bare, **145.55** with nylon and **120.00** with aluminium. The figure uses independent modal integration.

Native MAPDL confirms both base-X results. Aluminium reduced board motion relative to nylon, but raised peak stress across the PLA layers near a screw bearing from 17.16 to 24.78 MPa RMS. Both models reached cell contact during the full X history. In the 480-frame clip, only the nylon model crossed the gap (frames 253–270, counting from 1).